Agent Oriented Programming with Jason @JaCaMo

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ALTISSIMO 2014 — Bucharest

Outline of the course

- Introduction
- AOP – Agent Oriented Programming: *Jason*
- EOP – Environment Oriented Programming: CArtAgO (Alessandro Ricci)
- OOP – Organisation Oriented Programming: $\mathcal{M}$oise (Olivier Boissier)
- Conclusions
- Practical Exercise: a hands-on lab session!

Multi-Agent System (our perspective)

def...

An organisation of autonomous agents interacting together within a shared environment

- agents can be: software/hardware, coarse-grain/small-grain, heterogeneous/homogeneous, reactive/pro-active entities
- environment can be virtual/physical, passive/active, deterministic/non deterministic, ...
- interaction is the motor of dynamic in MAS. Interaction can be: direct/indirect between agents, interaction between agent and environment
- organisation can be pre-defined/emergent, static/adaptive, open/closed, ...

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Multi-Agent System (our perspective)

def...

An organisation of autonomous agents interacting together within a shared environment

MAS is not a simple set of agents

- agents can be: software/hardware, coarse-grain/small-grain, heterogeneous/homogeneous, reactive/pro-active entities
- environment can be virtual/physical, passive/active, deterministic/non deterministic, ...
- interaction is the motor of dynamic in MAS. Interaction can be: direct/indirect between agents, interaction between agent and environment
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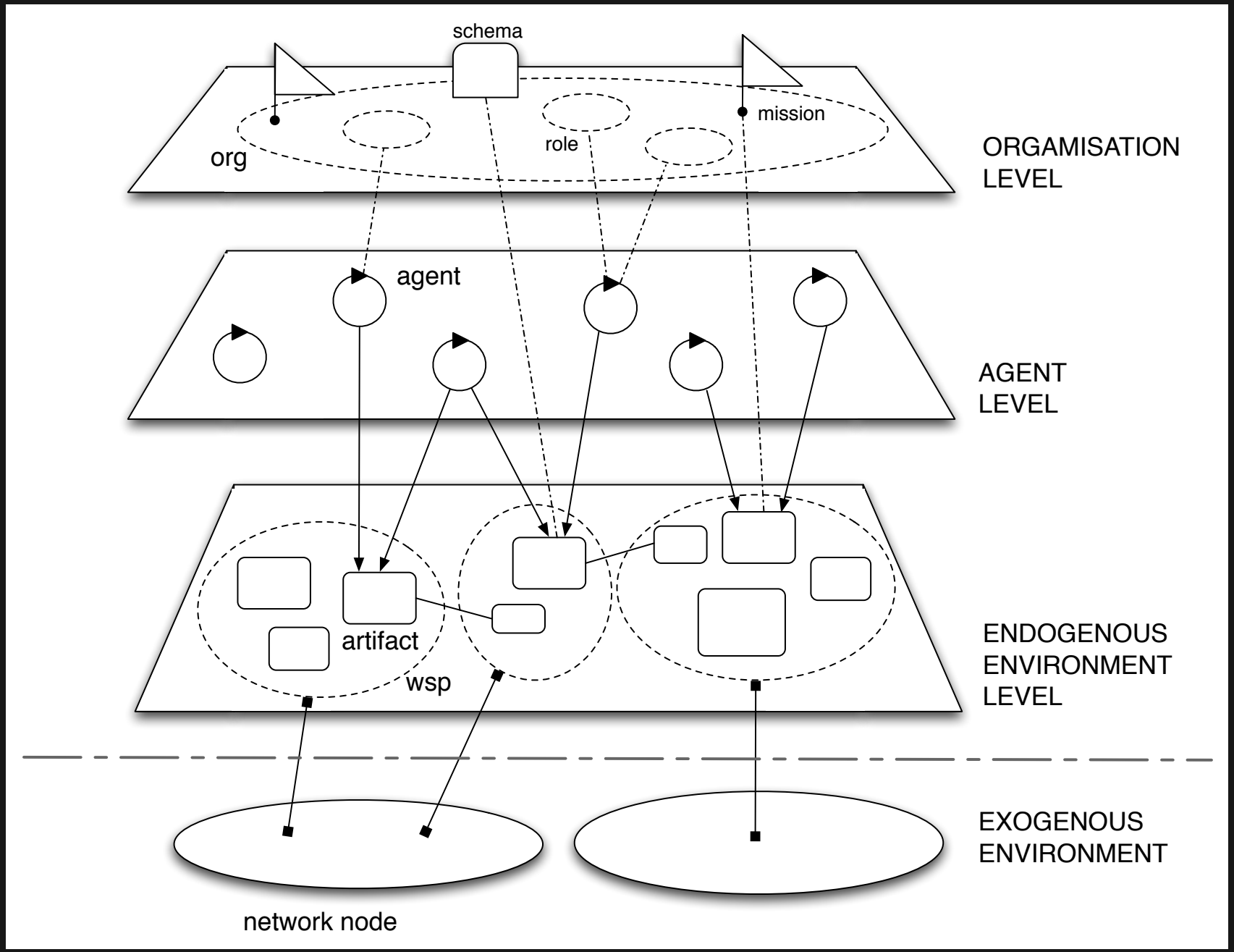
Levels in Multi-Agent Systems



Abstractions in Multi-Agent Systems

- Individual level
  - autonomy, situatedness
  - beliefs, desires, goals, intentions, plans
  - sense/reason/act, reactive/pro-active behaviour
- Environment level
  - resources and services that agents can access and control
  - sense/act
- Social level
  - cooperation, languages, protocols
- Organisation level
  - coordination, regulation patterns, norms, obligations, rights

Agent Oriented Programming

— **AOP** —

Literature I

Books: [Bordini et al., 2005a], [Bordini et al., 2009]

Proceedings: ProMAS, DALT, LADS, EMAS, ...

Surveys: [Bordini et al., 2006a], [Fisher et al., 2007] ...

Languages of historical importance: Agent0 [Shoham, 1993],
AgentSpeak(L) [Rao, 1996],
MetateM [Fisher, 2005],
3APL [Hindriks et al., 1997],
Golog [Giacomo et al., 2000]

Other prominent languages:
Jason [Bordini et al., 2007b],
Jadex [Pokahr et al., 2005], 2APL [Dastani, 2008],
GOAL [Hindriks, 2009], JACK [Winikoff, 2005],
JIAC, AgentFactory

But many others languages and platforms...

Some Languages and Platforms

Jason (Hübner, Bordini, ...); 3APL and 2APL (Dastani, van Riemsdijk, Meyer, Hindriks, ...); Jadex (Braubach, Pokahr); MetateM (Fisher, Guidini, Hirsch, ...); ConGoLog (Lesperance, Levesque, ... / Boutilier – DTGolog); Teamcore/ MTDP (Milind Tambe, ...); IMPACT (Subrahmanian, Kraus, Dix, Eiter); CLAIM (Amal El Fallah-Seghrouchni, ...); GOAL (Hindriks); BRAHMS (Sierhuis, ...); SemantiCore (Blois, ...); STAPLE (Kumar, Cohen, Huber); Go! (Clark, McCabe); Bach (John Lloyd, ...); MINERVA (Leite, ...); SOCS (Torroni, Stathis, Toni, ...); FLUX (Thielscher); JIAC (Hirsch, ...); JADE (Agostino Poggi, ...); JACK (AOS); Agentis (Agentis Software); Jackdaw (Calico Jack); ...

The State of Multi-Agent Programming

- Already the right way to implement MAS is to use an AOSE methodology (Prometheus, Gaia, Tropos, ...) and an MAS programming language!
- Many agent languages have efficient and stable interpreters — used extensively in teaching
- All have some programming tools (IDE, tracing of agents' mental attitudes, tracing of messages exchanged, etc.)
- Finally integrating with social aspects of MAS
- Growing user base

Agent Oriented Programming

Features

- Reacting to events $\times$ long-term goals
- Course of actions depends on circumstance
- Plan failure (dynamic environments)
- Social ability
- Combination of theoretical and practical reasoning

Agent Oriented Programming

Fundamentals

- Use of mentalistic notions and a societal view of computation [Shoham, 1993]
- Heavily influence by the BDI architecture and reactive planning systems [Bratman et al., 1988]

BDI architecture

[Wooldridge, 2009]

```
begin
    while true do
        p ← perception()
        B ← brf(B, p) ;                    // belief revision
        D ← options(B, I) ;                // desire revision
        I ← filter(B, D, I) ;              // deliberation
        execute(I) ;                       // means-end
end
```

BDI architecture

[Wooldridge, 2009]

```
while true do
    B ← brf(B, perception())
    D ← options(B, I)
    I ← filter(B, D, I)
    π ← plan(B, I, A)
    while π ≠ ∅ do
        execute( head(π) )
        π ← tail(π)
```

BDI architecture

[Wooldridge, 2009]

```
while true do
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BDI architecture

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```
while true do
    B ← brf(B, perception())
    D ← options(B, I)
    I ← filter(B, D, I)
    π ← plan(B, I, A)
    while π ≠ ∅ do
        execute( head(π) )
        π ← tail(π)
        B ← brf(B, perception())
        if ¬sound(π, I, B) then
            π ← plan(B, I, A)
```

revise commitment to plan – re-planning for context adaptation

BDI architecture

[Wooldridge, 2009]

```
while true do
    B ← brf(B, perception())
    D ← options(B, I)
    I ← filter(B, D, I)
    π ← plan(B, I, A)
    while π ≠ ∅ and ¬succeeded(I, B) and ¬impossible(I, B) do
        execute( head(π) )
        π ← tail(π)
        B ← brf(B, perception())
        if ¬sound(π, I, B) then
            π ← plan(B, I, A)
```

revise commitment to intentions – Single-Minded Commitment

BDI architecture

[Wooldridge, 2009]

```
while true do
    B ← brf(B, perception())
    D ← options(B, I)
    I ← filter(B, D, I)
    π ← plan(B, I, A)
    while π ≠ ∅ and ¬succeeded(I, B) and ¬impossible(I, B) do
        execute( head(π) )
        π ← tail(π)
        B ← brf(B, perception())
        if reconsider(I, B) then
            D ← options(B, I)
            I ← filter(B, D, I)
        if ¬sound(π, I, B) then
            π ← plan(B, I, A)
```

reconsider the intentions (not always!)

Jason

(let's go programming those nice concepts)

(BDI) Hello World

```
i_am(happy).                                    // B
!say(hello).                                    // D

+!say(X) : not i_am(sad) <- .print(X).          // I
```

Desires in Hello World

```
+i_am(happy) <- !say(hello).

+!say(X) : not i_am(sad) <- .print(X).
```

Hello World

source of beliefs

```
+i_am(happy)[source(A)]
    :  someone_who_knows_me_very_well(A)
    <- !say(hello).

+!say(X) : not i_am(sad) <- .print(X).
```

Hello World

plan selection

```
+is_happy(H)[source(A)]
    :  sincere(A) & .my_name(H)
    <- !say(hello).

+is_happy(H)
    :  not .my_name(H)
    <- !say(i_envy(H)).

+!say(X) : not i_am(sad) <- .print(X).
```

Hello World

intention revision

```
+is_happy(H)[source(A)]
    :  sincere(A) & .my_name(H)
   <- !say(hello).

+is_happy(H)
    :  not .my_name(H)
   <- !say(i_envy(H)).

+!say(X) : not i_am(sad) <- .print(X); !say(X).

-is_happy(H)
    :  .my_name(H)
   <- .drop_intention(say(hello)).
```

Hello World

intention revision

```
+is_happy(H)[source(A)]
    :  sincere(A) & .my_name(H)
   <- !say(hello).

+is_happy(H)
    :  not .my_name(H)
   <- !say(i_envy(H)).

+!say(X) : not i_am(sad) <- .print(X); !say(X).

-is_happy(H)
    :  .my_name(H)
   <- .drop_intention(say(hello)).
```

AgentSpeak

The foundational language for *Jason*

- Originally proposed by Rao [Rao, 1996]
- Programming language for BDI agents
- Elegant notation, based on logic programming
- Inspired by PRS (Georgeff & Lansky), dMARS (Kinny), and BDI Logics (Rao & Georgeff)
- Abstract programming language aimed at theoretical results

Jason

A practical implementation of a variant of AgentSpeak

- *Jason* implements the operational semantics of a variant of AgentSpeak
- Has various extensions aimed at a more practical programming language (e.g. definition of the MAS, communication, ...)
- Highly customised to simplify extension and experimentation
- Developed by Jomi F. Hübner, Rafael H. Bordini, and others

Main Language Constructs

Beliefs: represent the information available to an agent (e.g. about the environment or other agents)

Goals: represent states of affairs the agent wants to bring about

Plans: are recipes for action, representing the agent's know-how

Events: happen as consequence to changes in the agent's beliefs or goals

Intentions: plans instantiated to achieve some goal

Main Language Constructs and Runtime Structures

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Plans: are recipes for action, representing the agent's know-how

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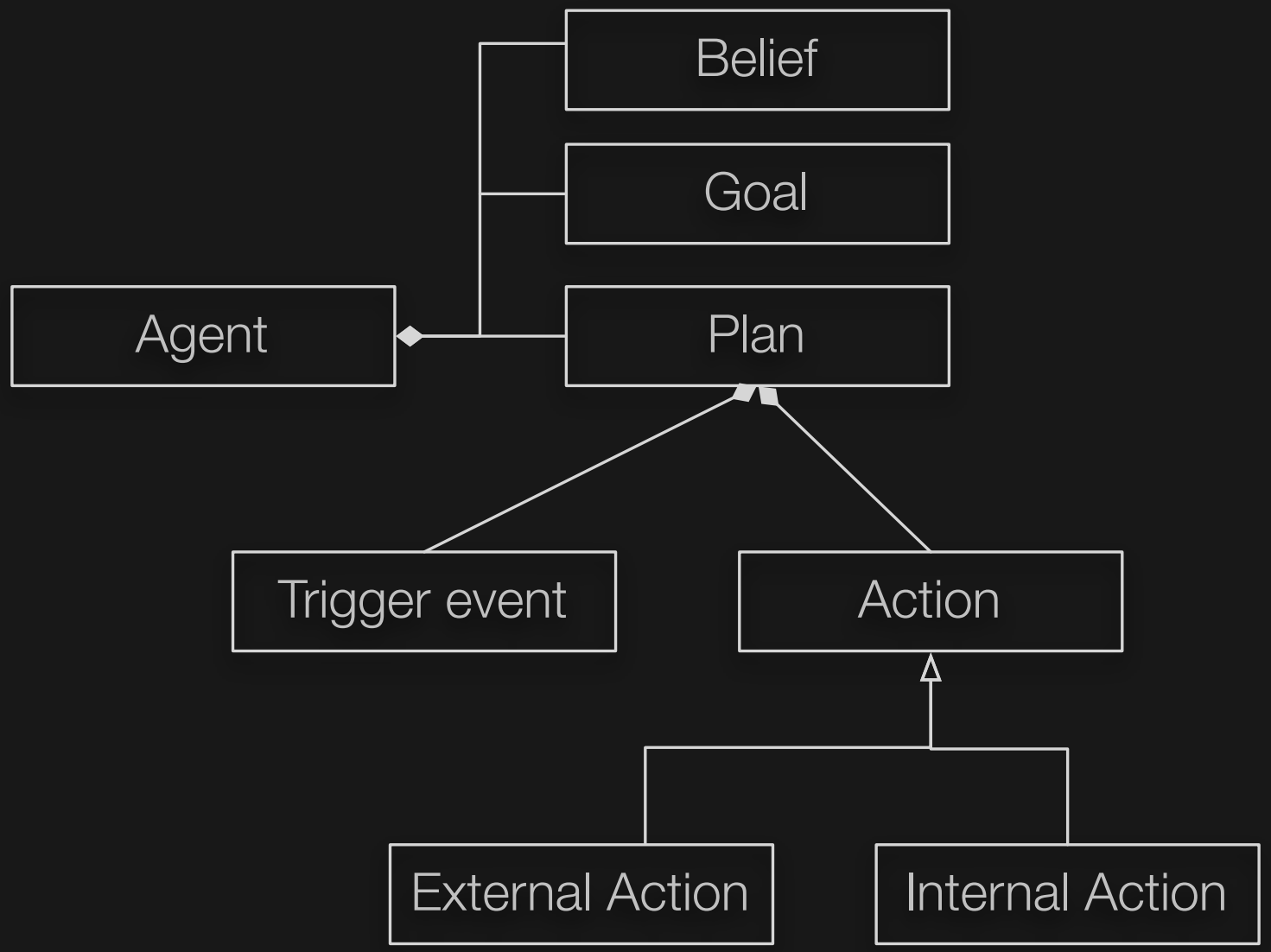
Agent meta-model

main constructors

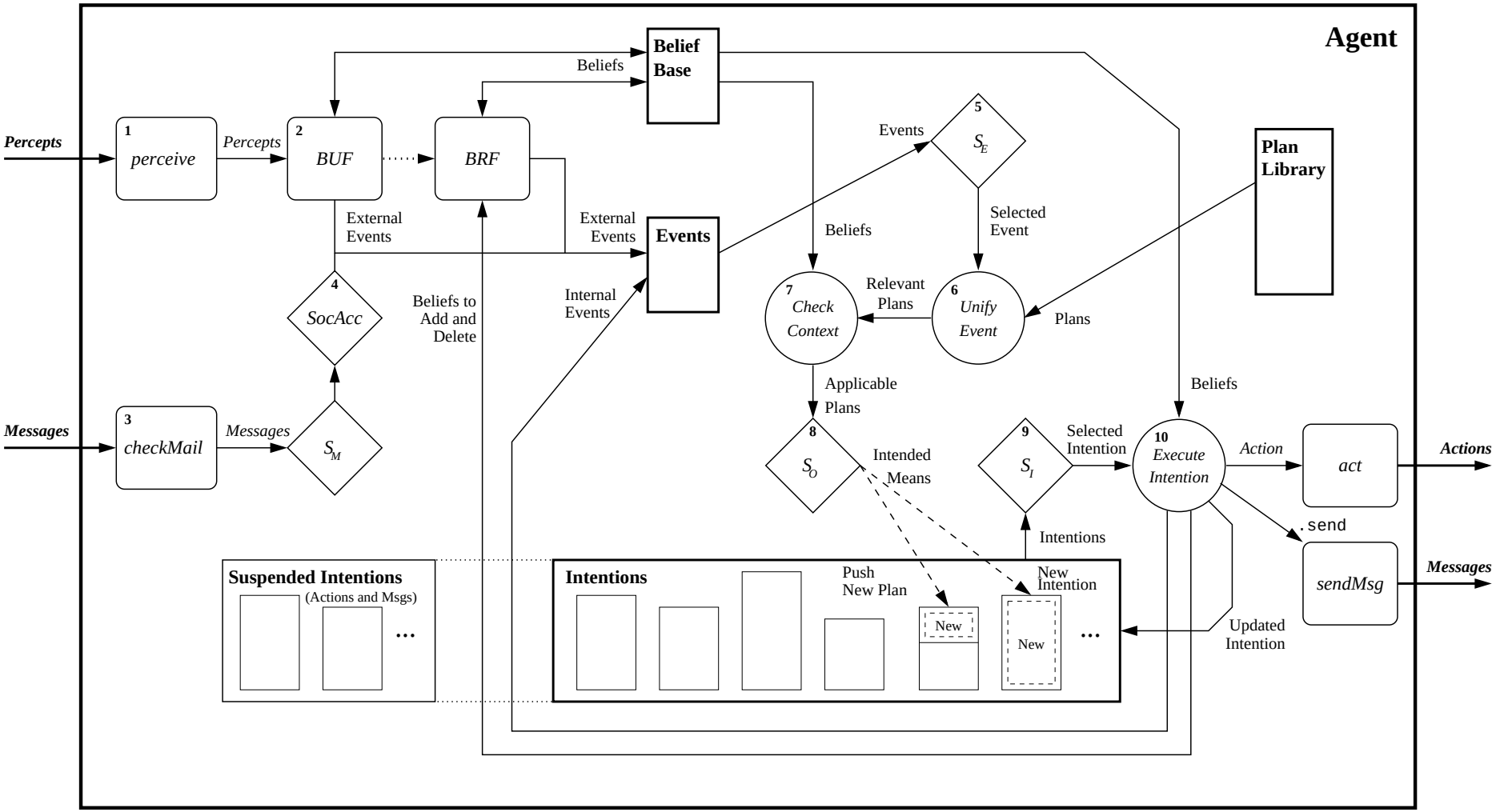


Basic Reasoning cycle

runtime interpreter

- perceive the environment and update belief base
- process new messages
- select event
- select relevant plans
- select applicable plans
- create/update intention
- select intention to execute
- execute one step of the selected intention

Jason Reasoning Cycle



Beliefs — Representation

Syntax

Beliefs are represented by annotated literals of first order logic

`functor(`$term_1$`, ..., `$term_n$`)[`$annot_1$`, ..., `$annot_m$`]`

Example (belief base of agent Tom)

```
red(box1)[source(percept)].
friend(bob,alice)[source(bob)].
lier(alice)[source(self),source(bob)].
~lier(bob)[source(self)].
```

Beliefs — Dynamics I

by perception

beliefs annotated with source(percept) are automatically updated accordingly to the perception of the agent

by intention

the plan operators + and - can be used to add and remove beliefs annotated with source(self) (mental notes)

```
+lier(alice); // adds lier(alice)[source(self)]
-lier(john); // removes lier(john)[source(self)]
```

Beliefs — Dynamics II

by communication

when an agent receives a tell message, the content is a new belief annotated with the sender of the message

```
.send(tom,tell,lier(alice)); // sent by bob
// adds lier(alice)[source(bob)] in Tom's BB
...
.send(tom,untell,lier(alice)); // sent by bob
// removes lier(alice)[source(bob)] from Tom's BB
```

Goals — Representation

Types of goals

- Achievement goal: goal to do
- Test goal: goal to know

Syntax

Goals have the same syntax as beliefs, but are prefixed by
! (achievement goal) or
? (test goal)

Example (Initial goal of agent Tom)

```
!write(book).
```

Goals — Dynamics I

by intention

the plan operators **!** and **?** can be used to add a new goal annotated with source(self)

```
...
// adds new achievement goal !write(book)[source(self)]
!write(book);

// adds new test goal ?publisher(P)[source(self)]
?publisher(P);
...
```

Goals — Dynamics II

by communication – achievement goal

when an agent receives an achieve message, the content is a new achievement goal annotated with the sender of the message

```
.send(tom,achieve,write(book)); // sent by Bob
// adds new goal write(book)[source(bob)] for Tom
...
.send(tom,unachieve,write(book)); // sent by Bob
// removes goal write(book)[source(bob)] for Tom
```

Goals — Dynamics III

by communication – test goal

when an agent receives an askOne or askAll message, the content is a new test goal annotated with the sender of the message

```
.send(tom,askOne,published(P),Answer); // sent by Bob
// adds new goal ?publisher(P)[source(bob)] for Tom
// the response of Tom will unify with Answer
```

Triggering Events — Representation

- Events happen as consequence to changes in the agent's beliefs or goals
- An agent reacts to events by executing plans
- Types of plan triggering events

+b	(belief addition)
-b	(belief deletion)
+!g	(achievement-goal addition)
-!g	(achievement-goal deletion)
+?g	(test-goal addition)
-?g	(test-goal deletion)

Plans — Representation

An AgentSpeak plan has the following general structure:

triggering_event : context <- body.

where:

- the triggering event denotes the events that the plan is meant to handle
- the context represent the circumstances in which the plan can be used
- the body is the course of action to be used to handle the event if the context is believed true at the time a plan is being chosen to handle the event

Plans — Operators for Plan **Context**

Boolean operators

&	(and)
|	(or)
not	(not)
=	(unification)
>, >=	(relational)
<, <=	(relational)
==	(equals)
\ ==	(different)

Arithmetic operators

+	(sum)
-	(subtraction)
*	(multiply)
/	(divide)
div	(divide – integer)
mod	(remainder)
**	(power)

Plans — Operators for Plan **Body**

```
+rain :  time_to_leave(T) & clock.now(H) & H >= T
   <- !g1;            // new sub-goal
      !!g2;           // new goal
      ?b(X);          // new test goal
      +b1(T-H);       // add mental note
      -b2(T-H);       // remove mental note
      -+b3(T*H);      // update mental note
      jia.get(X);     // internal action
      X > 10;         // constraint to carry on
      close(door);// external action
      !g3[hard_deadline(3000)].  // goal with deadline
```

Plans — Example

```
+green_patch(Rock)[source(percept)]
    :  not battery_charge(low)
    <- ?location(Rock,Coordinates);
       !at(Coordinates);
       !examine(Rock).

+!at(Coords)
    :  not at(Coords) & safe_path(Coords)
    <- move_towards(Coords);
       !at(Coords).
+!at(Coords)
    :  not at(Coords) & not safe_path(Coords)
    <- ...
+!at(Coords) :  at(Coords).
```

Plans — Dynamics

The plans that form the plan library of the agent come from

- initial plans defined by the programmer
- plans added dynamically and intentionally by
  - .add_plan
  - .remove_plan
- plans received from
  - tellHow messages
  - untellHow

A note about "Control"

Agents can control (manipulate) their own (and influence the others)

- beliefs
- goals
- plan

By doing so they control their behaviour

The developer provides initial values of these elements and thus also influence the behaviour of the agent

Other Language Features

Strong Negation

```
+!leave(home)
    :  ~raining
    <- open(curtains); ...

+!leave(home)
    :  not raining & not ~raining
    <- .send(mum,askOne,raining,Answer,3000); ...
```

Prolog-like Rules in the Belief Base

```
tall(X) :-
   woman(X) & height(X, H) & H > 1.70
   |
   man(X) & height(X, H) & H > 1.80.

likely_color(Obj,C) :-
   colour(Obj,C)[degOfCert(D1)] &
   not (colour(Obj,_)[degOfCert(D2)] & D2 > D1) &
   not ~colour(C,B).
```

Plan Annotations

- Like beliefs, plans can also have annotations, which go in the plan label
- Annotations contain meta-level information for the plan, which selection functions can take into consideration
- The annotations in an intended plan instance can be changed dynamically (e.g. to change intention priorities)
- There are some pre-defined plan annotations, e.g. to force a breakpoint at that plan or to make the whole plan execute atomically

Example (an annotated plan)

```
@myPlan[chance_of_success(0.3), usual_payoff(0.9),
        any_other_property]
+!g(X) : c(t) <- a(X).
```

Failure Handling: Contingency Plans

Example (an agent blindly committed to g)

```
+!g :  g.

+!g :  ...  <- ...  ?g.

-!g :  true <- !g.
```

Meta Programming

Example (an agent that asks for plans *on demand*)

```
-!G[error(no_relevant)] :  teacher(T)
   <- .send(T, askHow, { +!G }, Plans);
      .add_plan(Plans);
      !G.
```

in the event of a failure to achieve **any** *goal G due to no relevant plan, asks a teacher for plans to achieve G and then try G again*

- The failure event is annotated with the error type, line, source, ... error(no_relevant) means no plan in the agent's plan library to achieve G
- { +!G } is the syntax to enclose triggers/plans as terms

Internal Actions

- Unlike actions, internal actions do not change the environment
- Code to be executed as part of the agent reasoning cycle
- AgentSpeak is meant as a high-level language for the agent's practical reasoning and internal actions can be used for invoking legacy code elegantly

- Internal actions can be defined by the user in Java

```
libname.action_name(...)
```

Standard Internal Actions

- Standard (pre-defined) internal actions have an empty library name
  - `.print(`$term_1$`,`$term_2$`,...)`
  - `.union(`$list_1$`, `$list_2$`, `$list_3$`)`
  - `.my_name(`var`)`
  - `.send(`ag`,`$perf$`,`$literal$`)`
  - `.intend(`$literal$`)`
  - `.drop_intention(`$literal$`)`
- Many others available for: printing, sorting, list/string operations, manipulating the beliefs/annotations/plan library, creating agents, waiting/generating events, etc.

Jason × Java I

Consider a very simple robot with two goals:

- when a piece of gold is seen, go to it
- when battery is low, go charge it

Jason × Java II

Example (Java code – go to gold)

```
public class Robot extends Thread {
   boolean seeGold, lowBattery;
   public void run() {
      while (true) {
          while (!  seeGold) {
              a = randomDirection();
              doAction(go(a));

          }
          while (seeGold) {
              a = selectDirection();

              doAction(go(a));

} } }   }
```

Jason × Java III

Example (Java code – charge battery)

```
public class Robot extends Thread {
   boolean seeGold, lowBattery;
   public void run() {
      while (true) {
          while (!  seeGold) {
              a = randomDirection();
              doAction(go(a));
              if (lowBattery) charge();
          }
          while (seeGold) {
              a = selectDirection ();
              if (lowBattery) charge();
              doAction(go(a));
              if (lowBattery) charge();
} } }   }
```

Jason × Java IV

Example (*Jason* code)

```
direction(gold)   :- see(gold).
direction(random) :- not see(gold).

+!find(gold)                          // long term goal
   <- ?direction(A);
      go(A);
      !find(gold).
+battery(low)                         // reactivity
   <- !charge.

^!charge[state(started)]              // goal meta-events
   <- .suspend(find(gold)).
^!charge[state(finished)]
   <- .resume(find(gold)).
```

Jason × Prolog

- With the *Jason* extensions, nice separation of theoretical and practical reasoning
- BDI architecture allows
  - long-term goals (goal-based behaviour)
  - reacting to changes in a dynamic environment
  - handling multiple foci of attention (concurrency)
- Acting on an environment and a higher-level conception of a distributed system

Communication **Infrastructure**

Various communication and execution management infrastructures can be used with *Jason*:

Centralised:	all agents in the same machine, one thread by agent, very fast
Centralised (pool):	all agents in the same machine, fixed number of thread, allows thousands of agents
Jade:	distributed agents, FIPA-ACL
...	others defined by the user (e.g. AgentScape)

JaCaMo Configuration Language I

(beta version)

- Simple way of defining a multi-agent system

Example (MAS that uses JADE as infrastructure)

```
mas my_system {
   agent c3po
   agent r2d2
   agent bob
   workspace robots {  ....   }
}
mas-deployment {
   ag-instances: 1 c3po
                 1 r2d2 @ 10.10.1.2
                 10 bob
   ws-instances: robots
   platform: jade()
}
```

JaCaMo Configuration Language II

(beta version)

- Configuration of event handling, frequency of perception, user-defined settings, customisations, etc.

Example (MAS with customised agent)

```
agent bob {
     verbose: 2
     ag-class: MyAg
     ag-arch:  MyAgArch
     ag-bb-class: jason.bb.JDBCPersistentBB(
                     "org.hsqldb.jdbcDriver",
                     "jdbc:hsqldb:bookstore",
                     ...
}
```

Jason Customisations

- Agent class customisation:
  selectMessage, selectEvent, selectOption, selectIntetion, buf, brf, ...
- Agent architecture customisation:
  perceive, act, sendMsg, checkMail, ...
- Belief base customisation:
  add, remove, contains, ...
  - Example available with *Jason*: persistent belief base (in text files, in data bases, ...)

Tools

- Eclipse Plugin
- Mind Inspector
- Integration with
  - CArtAgO
  - $\mathcal{M}$oise
  - MADEM
  - Ontologies
  - ...
- More on `http://jason.sourceforge.net/wp/projects/`

Summary

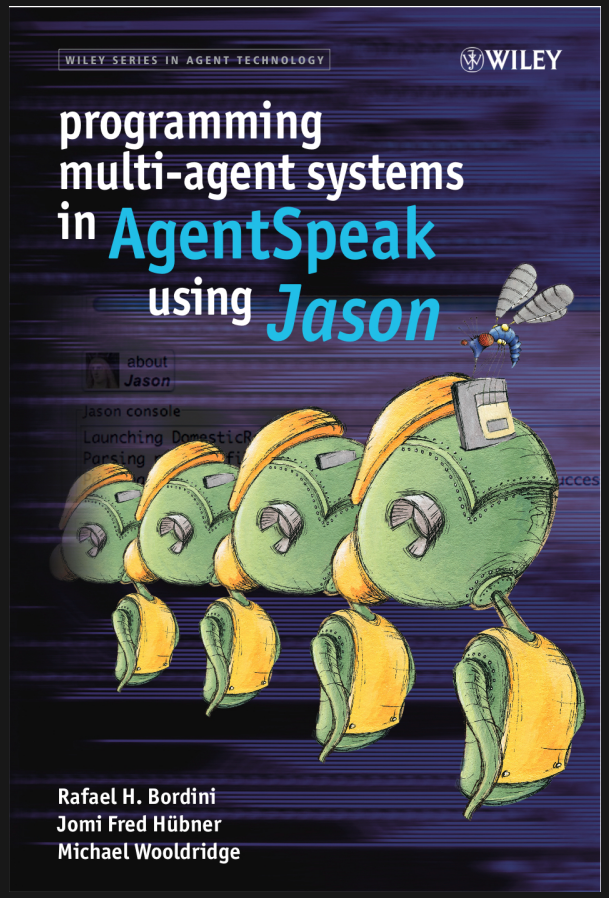
- AgentSpeak
  - Logic + BDI
  - Agent programming language
- *Jason*
  - AgentSpeak interpreter
  - Implements the operational semantics of AgentSpeak
  - Speech-act based communicaiton
  - Highly customisable
  - Useful tools
  - Open source
  - Open issues

Acknowledgements

- Many thanks to the
  - Various colleagues acknowledged/referenced throughout these slides
  - *Jason* users for helpful feedback
  - CNPq for supporting some of our current researh

Further Resources

- http://jason.sourceforge.net
- R.H. Bordini, J.F. Hübner, and M. Wooldrige
  Programming Multi-Agent Systems in AgentSpeak using *Jason*
  John Wiley & Sons, 2007.



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